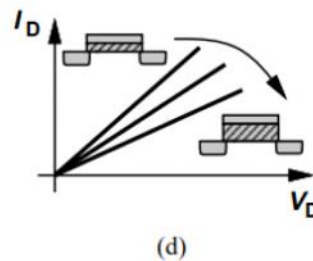
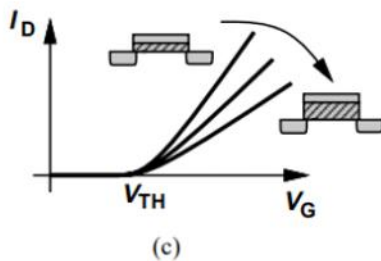
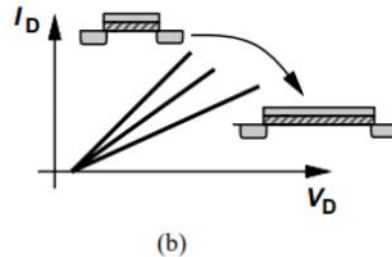
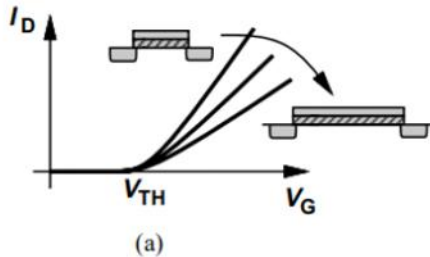


Question # 1 : [CLO-1]**[20]**

Considering the following four depiction which highlight different aspects of the MOSFET structure along with their impact on its IV characteristics. Label each depiction and highlight which structural feature's impact is shown on the IV plot.



a) Transfer curve:

Different effects of channel length.

- As length increases $\rightarrow$ on resistance increases.
- For $V_{GS} > V_{TH}$, drain current begins at lesser values L increases.
- It is preferable to minimize length to achieve large drain currents.

b) Output curve:

- Effects of channel length (L) I_D vs V_D .

increasing channel length impacts the IV:

- I_D exhibits a lesser slope as function of V_D
- higher on-resistance means less current delivered for the V_D
- As we increase L , curves get more compressed downwards.

c) Transfer Curve:

effects of (t_{ox}) I_D vs V_D :

- Structural features: Increasing gate oxide thickness (t_{ox}) impact of I_D :
- As t_{ox} increases: $C_{ox} = \epsilon_{ox}/t_{ox}$ decreases
- Curve shift downward as t_{ox} increases.

d) Output curve

• Effect of Oxide thickness I_D vs V_D :

- Increasing the oxide thickness.

Impact on I_D :-

- Higher $t_{ox} \rightarrow$ higher on-resistance $\rightarrow$ less drain current of drain voltage
- Output curves are suppressed downwards.

Question # 2 : [CLO-1]

[20]

The following tables express contradictory relationship between transconductance of a MOSFET and the overdrive voltage. Pick those relationships and clarify that contradiction in simple words.

$\frac{W}{L}$ Constant $V_{GS} - V_{TH}$ Variable	$\frac{W}{L}$ Variable $V_{GS} - V_{TH}$ Constant	$\frac{W}{L}$ Variable $V_{GS} - V_{TH}$ Constant
$g_m \propto \sqrt{I_D}$	$g_m \propto I_D$	$g_m \propto \sqrt{\frac{W}{L}}$
$g_m \propto V_{GS} - V_{TH}$	$g_m \propto \frac{W}{L}$	$g_m \propto \frac{1}{V_{GS} - V_{TH}}$

from basic MOSFETS drain current in saturation:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \quad \text{--- (a)}$$

So for g_m :

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) \quad \text{--- i}$$

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 \quad \text{--- i}$$

$$g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D} \quad \text{--- (ii)}$$

$$g_m = \frac{2 I_D}{V_{GS} - V_{TH}} \quad \text{--- (iii)}$$

• Contradiction:

• $g_m \propto (V_{GS} - V_{TH}) \rightarrow$ increases with overdrive, W/L constant.

• $g_m \propto \frac{1}{(V_{GS} - V_{TH})} \rightarrow$ decreases with overdrive; W/L variable

• As one says g_m increases and other says it decreases, so it's contradictory.

For $g_m \propto (V_{GS} - V_{TH})$

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})$$

here, $\frac{W}{L}$ is const

For $g_m \propto \frac{1}{(V_{GS} - V_{TH})}$

so,

$$g_m = \frac{2 I_D}{V_{GS} - V_{TH}}$$

$\therefore I_D$ is const, and W/L is allowed to vary. To keep I_D constant while increasing overdrive, W/L must reduce.

Question # 3 : [CLO1]**[20]**

The following table lists BJT properties, list corresponding MOSFET property in front of it.

Bipolar Transistor
Exponential Characteristic Active: $V_{CB} > 0$ Saturation: $V_{CB} < 0$ Finite Base Current Early Effect Diffusion Current -

Bipolar Transistor

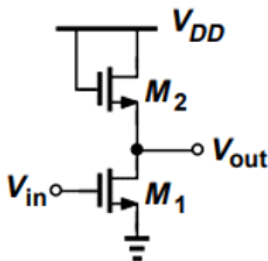
MOSFETS

- Exponential Characteristics
- Active: $V_{CB} > 0$
- Saturation: $V_{CB} < 0$
- finite base current
- Early effects
- Diffusion current

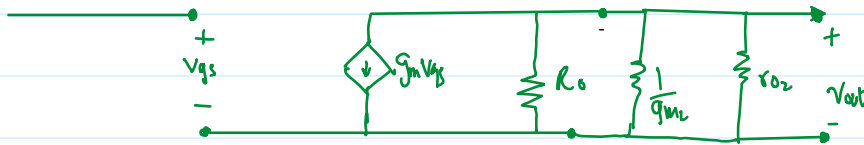
- Quadratic Character
- Saturation: $V_{DS} > V_{GS} - V_{TH}$ (NMOS)
- Triode: $V_{DS} < V_{GS} - V_{TH}$ (NMOS)
- Zero Base Current
- Drift current
- Voltage dependent resistor.

Question # 4 : [CLO-2]**[20]**

Using small signal model, calculate the expression for small signal gain of the following common-source stage with diode connected load. Assume there is no channel length modulation.



- first we will make small signal Model:



$$A_v = -g_{m1} \times r_{load}$$

$$A_v = -g_{m1} \times \frac{1}{g_{m2}} = -\frac{g_{m1}}{g_{m2}}$$

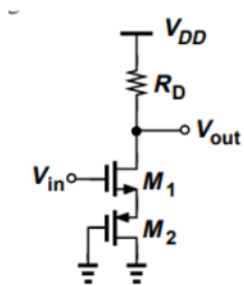
$$\therefore g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D}$$

$$A_v = \sqrt{\frac{(W/L)_1}{(W/L)_2}}$$

Question # 5 : [CLO-3]

[20]

Calculate the small signal gain of the following circuit:



Since M_2 is diode connected, it gives an impedance of

$$R_s = \frac{1}{g_{m2}}$$

Now the circuit becomes CS stage with source degeneration resistance = $\frac{1}{g_{m2}}$

$$A_v = \frac{-R_D}{\frac{1}{g_{m1}} + R_s}$$

$$\text{as } R_s = \frac{1}{g_{m2}}$$

$$A_v = \frac{-R_D}{1 + 1}$$

$$A_v = \frac{-R_D}{\frac{1}{g_{m1}} + \frac{1}{g_{m2}}}$$